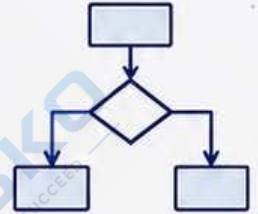


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Question Bank

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Core Java Programming

Unit I - Introduction to Java

- **Answers in one or two lines (1 mark questions)**

1. What is Java?
2. Who developed Java?
3. What was the original name of Java?
4. When was Java first released?
5. What is the main feature of Java that makes it platform independent?
6. Expand JDK.
7. Expand JRE.
8. Expand JVM.
9. What is JVM?
10. What is JDK?
11. What is JRE?
12. Name any two features of Java.
13. What is bytecode?
14. What is a Java keyword?
15. What is a data type?
16. Name the primitive data types in Java.
17. What is an operator?
18. What is the purpose of the `main()` method?
19. What is a command line argument?
20. What is the extension of a Java source file?
21. What is the extension of a Java bytecode file?
22. Which command is used to compile a Java program?
23. Which command is used to execute a Java program?
24. What is the default return type of the `main()` method?
25. Write the syntax of the `main()` method.

● **Short Answer Questions (3 marks questions)**

1. Explain the history and evolution of Java.
2. Explain any six features of Java.
3. Explain JDK, JRE, and JVM.
4. Differentiate between JDK, JRE, and JVM.
5. Explain the concept of Java bytecode.
6. Explain primitive data types in Java.
7. Explain non-primitive data types in Java.
8. Explain Java keywords with suitable examples.
9. Explain different types of operators in Java.
10. Explain arithmetic and relational operators in Java.
11. Explain logical and assignment operators in Java.
12. Explain the structure of a Java program.
13. Explain the purpose of the `main()` method in Java.
14. Explain command line arguments in Java with an example.
15. Explain the steps involved in compiling and executing a Java program.

● **Long Answer Questions (6 marks questions)**

1. Explain the history and evolution of Java in detail.
2. Explain the features of Java in detail with suitable examples.
3. Explain JDK, JRE, and JVM with a neat diagram.
4. Differentiate between JDK, JRE, and JVM and explain their functions.
5. Explain all primitive data types in Java with their size and range.
6. Explain different types of operators in Java with suitable examples.
7. Explain the complete structure of a Java program with a suitable example.
8. Explain Java command line arguments with a suitable program.
9. Explain the process of compiling and executing a Java program.
10. Write and explain simple Java programs demonstrating variables, data types, and operators.

Unit II - Decision Making and Looping

- **Answers in one or two lines (1 mark questions)**

1. What is decision making in Java?
2. What is an **if** statement?
3. What is an **if...else** statement?
4. What is a nested **if** statement?
5. What is an **else...if** ladder?
6. What is a **switch** statement?
7. What is looping?
8. Name the looping statements in Java.
9. What is a **while** loop?
10. What is a **do...while** loop?
11. What is a **for** loop?
12. What is a jump statement?
13. Name any three jump statements in Java.
14. What is the use of the **break** statement?
15. What is the use of the **continue** statement?
16. What is the difference between **while** and **do...while**?
17. Which loop executes at least once?
18. Which loop is generally used when the number of iterations is known?
19. What is the purpose of the **default** statement in **switch**?
20. What is the use of the **case** statement in Java?

- **Short Answer Questions (3 marks questions)**

1. Explain the simple **if** statement with syntax and example.
2. Explain the **if...else** statement with an example.
3. Explain nested **if...else** with an example.
4. Explain the **else...if** ladder with an example.
5. Explain the **switch** statement with syntax and examples.
6. Differentiate between **if...else** and **switch**.
7. Explain the **while** loop with an example.
8. Explain the **do...while** loop with an example.
9. Explain the **for** loop with an example.
10. Differentiate between **while** and **do...while** loops.
11. Differentiate between **while** and **for** loops.
12. Explain the **break** statement with an example.
13. Explain the **continue** statement with an example.
14. Explain jump statements in Java.
15. Explain nested loops with a suitable example.

- **Long Answer Questions (6 marks questions)**

1. Explain different decision-making statements in Java with suitable examples.
2. Explain **if**, **if...else**, nested **if...else**, and **else...if** ladder with syntax and programs.
3. Explain the **switch** statement in detail with syntax, rules, and a suitable program.
4. Differentiate between **if...else** ladder and **switch** statement with suitable examples.
5. Explain different looping statements in Java with suitable examples.
6. Explain **while**, **do...while**, and **for** loops with syntax and programs.
7. Differentiate between **while**, **do...while**, and **for** loops.
8. Explain jump statements **break**, **continue**, and **return** with suitable examples.
9. Explain nested loops in Java with a suitable program.
10. Write Java programs using decision-making and looping statements to solve suitable problems.

Unit III - Classes, Objects, Methods and Access Specifiers

- **Answers in one or two lines (1 mark questions)**

1. Define a class in Java.
2. Define an object.
3. What is a class member?
4. What is a data member?
5. What is a method?
6. What is object creation?
7. Which keyword is used to create an object in Java?
8. What is the **new** keyword?
9. What is the **this** keyword?
10. What is a constructor?
11. What is a default constructor?
12. What is a parameterized constructor?
13. What is constructor overloading?
14. What is method overloading?
15. What is an access specifier?
16. Name the access specifiers in Java.
17. What is the **public** access specifier?
18. What is the **private** access specifier?
19. What is the **protected** access specifier?
20. What is the purpose of the **this** keyword?
21. Can a constructor have a return type?
22. When is a constructor called?
23. Can constructors be overloaded?
24. What is garbage collection in Java?
25. Is **delete** a Java keyword?

- **Short Answer Questions (3 marks questions)**

1. Explain classes and objects in Java with an example.
2. Explain how to define a class in Java.
3. Explain fields and methods with suitable examples.
4. Explain object creation using the **new** keyword.
5. Explain how to access class members.
6. Explain method overloading with an example.
7. Explain constructors in Java.
8. Explain different types of constructors.
9. Explain default constructor with an example.
10. Explain parameterized constructor with an example.
11. Explain constructor overloading.
12. Explain the **this** keyword with an example.
13. Explain access specifiers in Java.
14. Differentiate between **public**, **private**, and **protected**.
15. Explain the importance of classes and objects in Java.

- **Long Answer Questions (6 marks questions)**

1. Explain classes and objects in Java with suitable examples and programs.
2. Explain the definition of a class, declaration of fields, methods, and creation of objects with a suitable program.
3. Explain method overloading with suitable examples.
4. Explain constructors and their types with suitable Java programs.
5. Explain default and parameterized constructors with examples.
6. Explain constructor overloading with a suitable Java program.
7. Explain **new** and **this** keywords with suitable examples.
8. Explain access specifiers **public**, **private**, and **protected** in detail.
9. Differentiate between method overloading and constructor overloading with examples.

Unit IV - Inheritance and Interfaces

- **Answers in one or two lines (1 mark questions)**

1. Define Inheritance.
2. What is the purpose of inheritance in Java?
3. What is a superclass?
4. What is a subclass?
5. Which keyword is used to inherit a class in Java?
6. What is the use of the **extends** keyword?
7. What is the **super** keyword?
8. Name the types of inheritance supported through classes in Java.
9. Define Single Inheritance.
10. Define Multilevel Inheritance.
11. Define Hierarchical Inheritance.
12. Does Java support multiple inheritance through classes?
13. What is a **final** variable?
14. What is a **final** method?
15. What is a **final** class?
16. Define an abstract method.
17. Define an abstract class.
18. What is an interface?
19. Which keyword is used to define an interface?
20. Which keyword is used to implement an interface?
21. Which keyword is used to extend an interface?
22. Can an interface be instantiated?
23. Can an abstract class be instantiated?
24. What is interface implementation?
25. What is the difference between **extends** and **implements**?

- **Short Answer Questions (3 marks questions)**

1. Explain inheritance in Java with an example.
2. Explain the **extends** keyword with a suitable example.
3. Explain the **super** keyword with an example.
4. Explain Single Inheritance.
5. Explain Multilevel Inheritance.
6. Explain Hierarchical Inheritance.
7. Explain the use of the **final** variable.
8. Explain the use of the **final** method.
9. Explain the use of the **final** class.
10. Explain abstract methods with an example.
11. Explain abstract classes with an example.
12. Explain interfaces in Java.
13. Explain how to define an interface.
14. Explain how an interface is implemented by a class.
15. Differentiate between abstract class and interface.
16. Differentiate between **extends** and **implements**.
17. Explain how one interface can extend another interface.
18. Explain the advantages of inheritance.
19. Explain the advantages of interfaces.
20. Explain the different types of inheritance in Java.

- **Long Answer Questions (6 marks questions)**

1. Explain inheritance in Java and its different types with suitable diagrams and programs.
2. Explain Single, Multilevel, and Hierarchical Inheritance with suitable Java programs.
3. Explain the **super** and **extends** keywords with suitable examples.
4. Explain **final** variables, **final** methods, and **final** classes in detail.
5. Explain abstract methods and abstract classes with suitable Java programs.
6. Explain interfaces, their features, and their importance in Java.
7. Explain how to define, extend, and implement interfaces with suitable programs.
8. Differentiate between abstract classes and interfaces with suitable examples.

9. Explain the role of interfaces in achieving multiple inheritance in Java.
10. Write a Java program demonstrating inheritance and interface implementation.